

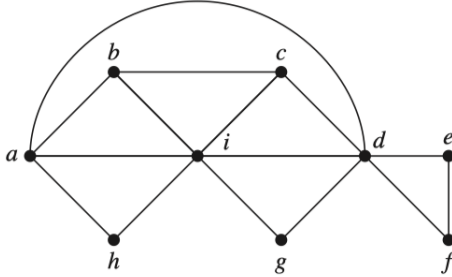
Exercise Sheet 14

Discrete Mathematics by Hongfei Fu, 2023.11.27

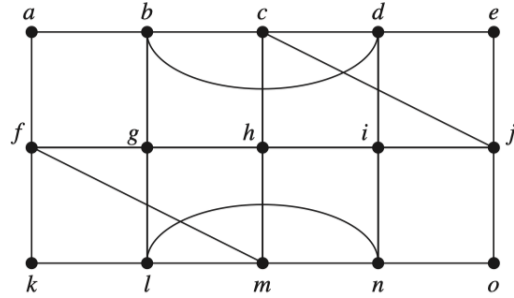
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1. ([R], Page 704, Exercise 6, 8) In Exercises 6, 8 determine whether the given graph has an Euler circuit. Construct such a circuit when one exists. If no Euler circuit exists, determine whether the graph has an Euler path and construct such a path if one exists.

6.



8.



Answer Area:

6. There isn't an Euler circuit, but there is an Euler path:

b, ba, a, ah, h, hi, i, ia, a, ad, d, dc, c, ci, i, ig, g, gd, d, de, e, ef, f, fd, d, di, i, ib, b, bc, c

8. There is an Euler circuit:

f, fm, m, ml, l, lk, k, kf, f, fg, g, gl, l, ln, n, nm, m, mh, h, hg, g, gb, b, bc, c, ch, h, hi, i, in, n, no, o, oj, j, jc, c, cd, d, di, i, ij, j, je, e, ed, d, db, b, ba, a, af

2. ([R], Page 704, Exercise 16, **2pt**) Show that a directed multigraph having no isolated vertices (i.e., vertices with zero in- and out-degree or with only self-loops) has an Euler circuit if and only if the graph is weakly connected and the in-degree and out-degree of each vertex are equal.

Note1: The definition of **weakly connected** is as follows:

Definition 1. A directed graph is **weakly connected** if there is a path between every two vertices in the underlying undirected graph (obtained by forgetting the direction of the edges in the original directed graph).

Note2: For a directed multigraph G , an Euler circuit in G is a simple circuit containing every edge of G , which means every edge is visited exactly once.

Answer Area:

- If a directed multigraph $G = (V, E)$ having no isolated vertices has an Euler circuit.
 $\therefore$ For every vertex in the circuit, if there is an edge in, there will be an edge out.
 $\therefore$ The in-degree and out-degree of each vertex are equal.
 $\therefore$ There is an Euler circuit. And Euler circuit contains every edge in the graph. That is, the circuit contains every vertex in the graph.
 $\therefore$ There is a path between every two vertices in the underlying undirected graph, which is a part of the circuit.
- If the graph $G = (V, E)$ is weakly connected and the in-degree and out-degree of each vertex are equal.

* Base step:

$|E| = 0$. Then G is not weakly connected provided that there isn't a path between every two vertices in the underlying undirected graph, so that the implication holds immediately.

* Inductive step:

Assuming that every graph whose number of edge is less than $|E|$ and the graph is weakly connected and the in-degree and out-degree of each vertex are equal, then the graph has an Euler circuit.

$\therefore$ The in-degree and out-degree of each vertex are equal.

$\therefore$ Choose an arbitrary vertex v_0 , there is an edge from v_0 to another vertex v_1 , then there is an edge from v_1 to another vertex $v_2, \dots$, repeat the process. Since the vertices in the graph is finite, the path must end up reaching a vertex v_i that has been reached before. Therefore there exists a circuit C from v_i to v_i that visits each vertex at most once (except for the start and end vertex). Let F be the set of edges appearing in C .

$\therefore$ Then each weakly connected component H of $G - F$ has a smaller number of edges, and the in-degree and out-degree of each vertex in H are equal since we delete the an edge in the vertex and an edge out of the vertex at the same time. For every such H , we have by the induction hypothesis that if H contains at least two vertices, then the weakly connected component H has an Euler circuit. Hence, we can construct an Euler circuit for G by traversing the vertices in C while finishing the Euler circuit of each connected component H having at least two vertices.

Therefore, G has an Euler circuit.

Therefore, a directed multigraph having no isolated vertices (i.e., vertices with zero in- and out-degree or with only self-loops) has an Euler circuit if and only if the graph is weakly connected and the in-degree and out-degree of each vertex are equal.